Mariya Savinov

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https://mariyasavinov.github.io/

Employment

NSF-Simons National Institute for Theory and Mathematics in Biology
NITMB Fellow Sept 2025 - *present*

University of Chicago
Chicago Fellow, Biological Sciences Division Sept 2025 - *present*

Education

Courant Institute of Mathematical Sciences, New York University
Ph.D. in Mathematics 2020 - 2025

- Advisor: Prof. Alex Mogilner
- Thesis: Biophysical modeling and simulation of contractile actomyosin dynamics

University of Pittsburgh
B.Phil. in Mathematics and Physics, with honors 2016 - 2020

- Advisor: Prof. Bard Ermentrout
- Honors Thesis: Synchronization and locking in oscillators with flexible periods

Research Interests

mathematical biology, collective dynamics, cell biology, mechanics, numerical simulation, data-driven methods, morphogenesis, modeling, dynamical systems, nonlinear phenomena

In preparation

8. Constrained autocatalytic assembly drives a transition from coexistence to selection in actin network competition
M. Savinov, C. Guérin, A. N'Diaye, L. Gressin, A. Mogilner, M. Théry, L. Blanchoin, A. Colin

7. Tumor cluster co-attraction is driven by the ECM
J. Garcia*, **M. Savinov***, A. Mogilner, C. Carmona-Fontaine

Publications

6. Simple cell motility on adhesive gradients
A. Mogilner and **M. Savinov**, *Nature Physics (News & Views)* vol. 21, pg. 1524-1525. doi: 10.1038/s41567-025-03010-8.

5. A model for contractile stress fibers embedded in bulk actomyosin networks
M. Savinov, C.S. Peskin, and A. Mogilner. *Journal of Mathematical Biology* vol. 91, no. 14, June 2025, doi: 10.1007/s00285-025-02245-6.

4. Size-dependent transition from steady contraction to waves in actomyosin networks with turnover
A. Krishna, **M. Savinov**, N. Ierushalmi, A. Mogilner, and K. Keren, *Nature Physics*, Jan 2024, vol. 20, pg. 123–134. doi: 10.1038/s41567-023-02271-5.

3. Friction patterns guide actin network contraction **equally contributing first authors*
A. Colin*, M. Orhant-Prioux*, C. Guérin*, **M. Savinov***, I. Scarfone, A. Roux, E.M. De La Cruz, A. Mogilner, M. Théry, and L. Blanchoin, *PNAS*, Sept 2023, vol. 120, no. 39, p. e2300416120, Sep. 2023, doi: 10.1073/pnas.2300416120.

2. Crawling, waving, inch worming, dilating, and pivoting mechanics of migrating cells: Lessons from Ken Jacobson
A. Mogilner, **M. Savinov**, *Biophysical J.*, Mar 2023, doi: 10.1016/j.bpj.2023.03.023

1. Synchronization and locking in oscillators with flexible periods
M. Savinov, D. Swigon, and B. Ermentrout, *Chaos*, vol. 31, no. 3, p. 033143, Mar. 2021, doi: 10.1063/5.0021836.

Awards and Honors

NSF Mathematical Sciences Postdoctoral Research Fellowship 2025
Awarded support: $190,000, ***declined***
Project Title: "Mechanosensitive assembly of yeast actin cables"

SIAM Travel Award, Society for Industrial and Applied Mathematics 2024
In support of attendance at SIAM Life Sciences 2024, $650

BPS Travel Award, Biophysical Society 2024
In support of attendance at BPS Annual Meeting 2024, $200

Paul Garabedian Fellowship, NYU Courant 2023
Awarded to an outstanding PhD student

ASCB Travel Grant, American Society for Cell Biology 2023
In support of attendance at Cell Bio 2023, $500

Martin & Sarah Leibowitz Graduate Prize for Quantitative Biology, Courant 2023
Awarded for academic excellence within the field of Quantitative Biology, $2500

SIAM Travel Award, Society for Industrial and Applied Mathematics 2023
In support of attendance at SIAM Dynamical Systems 2023, $650

Bella Manel Prize, NYU Courant 2022
Awarded for excellence and promise in mathematics on the graduate level, $500

Dean's Doctoral Fellowship, NYU 2020-2025

NSF Graduate Research Fellowship 2020-2025

Goldwater Scholarship 2019

DAAD Research Internships in Science and Engineering Scholarship 2019
Advisors: Prof. Pawel Romanczuk and Dr. Winnie Poel (Humboldt University of Berlin)

Sigma Pi Sigma (ΣΠΣ) **Induction,** American Institute of Physics 2019

Brackenridge Research Fellowship, PITT Honors College 2018

Koehler Academic Achievement Award, PITT Dept. of Physics and Astronomy 2018

NSF-REU in Extremal Graph Theory and Dynamical Systems 2017
Advisor: Prof. Elizabeth Cherry (Rochester Institute of Technology)

Presentations

JMM 2027 | *Invited AMS Special Session Presentation Jan 2027
- AMS Special Session on Collective Behavior, Decisions, and Dynamics

IIT Seminar in Applied Mathematics and Statistics | *Invited Nov 2026

Northwestern Complex Seminar Series | *Invited Nov 2026

SIAM Great Lakes 2026 | *Invited Minisymposium Presentation Sept 2026
- Session: Modeling and Data-Driven Approaches to Complex Biological Systems

SIAM Life Sciences 2026 | Minisymposium Organizer July 2026
- Session: Computational and Mathematical Approaches to Cytoskeletal Dynamics

Lake Conference: Modeling Life from Cells to Tissues | Poster April 2026

NASC RTG Ranch Retreat | *Invited Talk April 2026
- Hosted by Rice University Numerical Mathematics and Scientific Computing RTG

Presentations (cont.)

SMB Annual Meeting 2025 | *Invited Minisymposium Presentation — July 2025
- Session: Mechanistic modeling from inter- to intra-cellular phenomena

NYU Biology Department Annual Symposium | *Invited Trainee Speaker — June 2025

ASCB/EMBO 2024: Cell Bio | Subgroup Talk — Dec 2024
- Session: Theoretical Mechanobiology from Single Cell to Multicellular Level

UPenn Center for Mathematical Biology: MathBio Seminar | *Invited Talk — Oct 2024

ICERM: QCAM | Contributed Talk — June 2024
- Workshop: Queer in Computational and Applied Mathematics

SIAM Life Sciences 2024 | *Invited Minisymposium Presentation — June 2024
- Session: Data-driven Modeling of Spatial Subcellular Processes

PITT AWM Student Seminar | *Invited Talk — March 2024
- Talk recording can be found here

Biophysical Society Annual Meeting | Flash Talk (2598) and Poster (P3023) — Feb 2024

ASCB/EMBO 2023: Cell Bio | Subgroup Talk — Dec 2023
- Session: Building the Cell

SIAM Dynamical Systems 2023 | *Invited Minisymposium Presentation — May 2023
- Session: Self-organized pattenrs in cells: from omlecular to continuum mechanisms

ASCB/EMBO 2022: Cell Bio | Poster (P2438) — Dec 2022

NIMBioS Undergraduate Research Conference | Oral Presentation — Oct 2018

Professional, Outreach, and Service Activities

Peer reviewer | *Biophys J*, *PLOS Comp Bio*, *SIALS*, *eLife*, *Biophys Rep*

NITMB Topical Workshop | Organizer — March 22-24, 2027
- **Curiosity-Driven Dialogue and Collaboration Between Experiment and Theory**
- A workshop aimed at inspiring curiosity and revealing avenues of collaboration across the experiment-theory spectrum.
- Also held in Feb 3-5, 2026

UIC Young Scholars Program 2026 | Lecturer — July 21, 2026
- UIC's free summer math program for Chicago-area high schoolers.
- Guest lecture on developing a research career in mathematical biology, with project examples (60 minutes)

NITMB Work-in-Progress Meeting | Seminar Organizer — 2025-2026
- Co-organizing the schedule and moderating sessions for this seminar on in-progress NITMB research at various stages of completion in an informal setting to spark discussion, get feedback, and potentially form new collaborations.

Mathematical and Statistical Modeling of Complex Systems Workshop | Speaker
- Northeastern Illinois University's annual workshop for undergraduate students
- Contributed talk on Nov 20, 2025: "Resource competition modeling of actin-propelled beads" (30 minutes).

Modeling and Simulation Group Meeting | Seminar Organizer — 2023-2024 AY
- Co-organized the schedule and moderated sessions for this seminar in applied mathematics for graduate students, postdocs, and faculty at the Courant Institute. Seminar featured research talks, figure-making tutorials, and discussions of open problems and research challenges.

Professional, Outreach, and Service Activities (cont.)

cSplash 2024 | Lecturer — April 28, 2024
- NYU Courant's annual math-outreach lecture series for high schoolers.
- Contributed lecture "A crash course in modeling population dynamics" (40 minutes)

Courant PhD Student Survey | Lead student organizer — fall 2022, 2023
- Facilitated the design and analysis of an internal PhD student survey, with the intention to identify Courant student needs and concerns to department leadership. Roughly two-thirds of PhD students responded both years.

Courant DEI Reading Group | Organizer — 2021-2024
- Organizing and leading discussions on Diversity, Equity, and Inclusion issues in the context of Mathematics education and academia.

TESSA Talks | Founder and lead student organizer — 2018-2020
- Technology, Education, Science, Society, and Art discussion forum at Honors College

Mentoring

Undergraduate students (2)

Clara Hinds | Independent Study — spring 2026 - *present*
- Co-mentored with Prof. Jasmine Nirody
- Discovering the mechanism by which *Pseudomonas aeruginosa* biofilm microarchitecture drives changes in metabolic activity

Helen Tran | NITMB REU — summer 2026
- Co-mentored with Prof. Jasmine Nirody
- Modeling *Vibrio fischeri* aggregation dynamics in viscous mixing flows

Graduate students (3)

Anna Tang | MD-PhD Research rotation — summer 2025
- Co-mentored with Prof. Alex Mogilner
- Characterizing *Ciona* migrating cell shapes from experimental data using spherical harmonics

Rongjie Zhu | MS math thesis — 2024-2025 AY
- Co-advised with Prof. Alex Mogilner
- Mathematical modeling and simulation of collective 2-cell motility in *Ciona*

Tongtong Liu | MS math thesis — 2024-2025 AY
- Co-advised with Prof. Alex Mogilner
- Mathematical modeling and simulation of *in vitro* actomyosin dynamics

Teaching

MATH 365-0: Computational Methods in Mathematics — (upcoming) spring 2027
Instructor of Record — Northwestern University
- An introduction to computational methods in mathematics to analyze and interpret a variety of mathematical phenomena.

Teaching assistant for MATH-UA 258: Honors Numerical Analysis — spring 2025
Course instructor: Prof. Alan Kaptanoglu — New York University
- Designed weekly recitation section worksheets and held weekly office hours. Course covers the analysis of numerical algorithms ubiquitously used to solve problems throughout mathematics, physics, engineering, finance, and the life sciences.

Teaching assistant for MATH-UA 251: Mathematical Modeling — 2023-2024 AY
Course instructors: Prof. Aaditya Rangan, Prof. Alex Mogilner — New York University
- Designed recitation section materials and wrote weekly quizzes, with weekly office hours. Courses focused on the formulation and analysis of mathematical models for problems in biology, physics, economics, etc.

Teaching (cont.)

Course assistant for MATH-GA 2020: Graduate Numerical Methods II spring 2023
Course instructor: Prof. Aleksandar Donev New York University

- Ran weekly office hours and graded computational assignments, as well as assisted in creation of an example suite of ODE solvers with object-oriented design, responsible for designing the MATLAB codes (available on github.)
- On 02/28, taught 1h50min course lecture on "Absolute Stability and Stiffness"

Substitute Lecturer for MATH-UA 252: Numerical Analysis spring 2023
Course instructor: Dr. Samuel Potter New York University

- On 4/18 and 4/20, taught two 1h15min lectures introducing numerical methods for ODEs at the undergraduate level. Lecture notes can be found on my webpage.

Skills

Computing – advanced: `MATLAB, Python, LaTeX`

Computing – basic: `C, CUDA, MPI`

Russian, heritage speaker